

Enkh Baterdene

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Professional Summary

Journey .NET Application Developer with 7+ years of progressive experience at the Washington State Department of Labor & Industries, serving as technical lead for enterprise application development, architecture alignment, and cross-functional system integration. Deep expertise in C#/.NET, T-SQL stored procedures, WCF/REST web services, and secure Microsoft SDL/SDLC practices. Proven record of mentoring co-developers, sharing knowledge through pair programming and code reviews, enforcing coding standards, and delivering mission-critical solutions across authentication, ETL automation, secure financial transactions, and CI/CD pipeline modernization. Certified Scrum Master. Currently pursuing an M.S. in Computer Science with an AI/ML focus.

Core Technical Skills

Area	Technologies
Languages & Frameworks	C# / .NET Core, ASP.NET MVC / WCF, Python, Java, VB.NET
Database & Data	MS SQL Server, T-SQL, Stored Procedures, Entity Framework, ETL / SSIS
Web & Frontend	JavaScript, jQuery, HTML5, CSS3, Bootstrap, Node.js, Angular, Vue, React
APIs & Services	REST API, WCF, Web API, SOAP, JWT / Bearer Auth, OAuth2 / OpenID Connect
DevOps & Tooling	Git, TFS, Azure DevOps CI/CD, PowerShell, CMD / Bash, Docker, Visual Studio (Code)
Testing & QA	TDD, Cypress (E2E), NUnit, xUnit, Postman, SoapUI, Swagger
Methodology	Agile / Scrum, Certified Scrum Master (Scrum Alliance), Pair Programming
Architecture	MVC, Singleton, DI, Factory, Microservices, N-Tier, Design Patterns

Professional Experience

IT Application Developer – Journey (Technical Lead)

Washington State Department of Labor & Industries — April 2019 – Present

- Technical Lead & Architecture:** Serve as technical lead on an enterprise .NET application team, guiding architecture alignment, enforcing coding standards, and providing oversight across the full SDLC — design, development, deployment, and ongoing maintenance.
- Enterprise .NET & SQL Development:** Design, develop, and maintain high-impact C#/.NET web applications and WCF web services, applying MVC, N-Tier, and Singleton design patterns; author and maintain complex MS SQL stored procedures supporting rapid business-rule and feature changes.
- Modern .NET Development:** Build .NET Core Web API and Blazor applications on personal time to stay current with modern Microsoft platform capabilities and to inform architecture decisions for future modernization efforts.

- **Federated Identity Integration:** Integrated customer-facing applications with enterprise federated identity providers to enable secure external authentication and streamline cross-system portal interactions.
- **Secure API Development:** Engineered a secure .NET Web API client with JWT / Bearer token authentication for validating sensitive financial transactions, implementing role-based access control and data encryption.
- **ETL Automation:** Architected and deployed .NET console applications to automate scheduled bulk ETL operations from secure file-transfer sources, improving data-pipeline efficiency and reliability.
- **Mission-Critical Issue Resolution:** Resolved a high-precision arithmetic defect in financial calculations affecting downstream applications through deep root-cause analysis across interdependent systems.
- **CI/CD & DevOps Modernization:** Migrated 12+ console applications from TFS to Git and established full CI/CD pipelines through all environments to production. Currently coordinating with the DevOps team to migrate remaining web services and UI applications to Azure Git repositories with CI/CD support.
- **Test Automation:** Built automated Cypress E2E UI tests and .NET WCF integration tests to improve release confidence and reduce regression risk.
- **Stakeholder Communication & Reporting:** Collaborate with architects, managers, and cross-functional teams to deliver timely status reports, impact analysis, and technical documentation, including support for impact assessments and third-party software evaluations.
- **Mentorship & Agile Leadership:** Mentor co-developers, conduct in-depth code reviews, facilitate Agile ceremonies (standups, sprint planning, retrospectives), and coordinate with mainframe, DevOps, and network operations teams.
- **Certified Scrum Master** (Scrum Alliance, April 2024).

Software Engineer – Full Stack Java

Cognizant — *November 2018 – April 2019*

- Built microservices applications using Spring Boot, RESTful services, Spring Cloud, Netflix OSS (Eureka, Zuul, Feign), and Pivotal Cloud Foundry with CI/CD via Jenkins.
- Enabled SSO security with OpenID Connect, OAuth2, and JWT tokens via Keycloak across distributed services.
- Followed TDD, producing automated unit, integration, and smoke tests in Angular and Spring Boot.

Software Engineer – Full Stack Java

AT&T (via Infosys) — *February 2018 – November 2018*

- Contributed to enterprise application design, development, testing, debugging, documentation, and deployment for large-scale telecommunications systems.

Academic Projects — M.S. Computer Science, City University of Seattle

Serverless E-Commerce Backend Security: A Comparative Case Study

CS 547 – Secure Systems & Programs

- Architected a secure serverless application in Python and .NET, applying the Microsoft Secure Development Lifecycle (SDL) to integrate security and privacy from the foundational design phase.

- Conducted comprehensive threat modeling using the STRIDE framework and the Microsoft Threat Modeling Tool to systematically identify, categorize, and document potential architectural vulnerabilities.
- Mitigated critical OWASP Top 10 vulnerabilities (including SQL Injection and XSS) by applying quantitative risk-assessment scoring (DREAD / CVSS) to prioritize ethical-hacking efforts and streamline remediation.
- Engineered backend security controls implementing advanced authentication, Role-Based Access Control (RBAC), and cryptographic standards to safeguard sensitive data in transit and at rest.

Predictive Modeling & Neural Network Development

CS 620 – Machine Learning & Deep Learning

- Designed and trained ML models in Python with scikit-learn, TensorFlow, and PyTorch to solve complex classification and regression problems, applying feature engineering, normalization, and EDA on large-scale datasets.
- Implemented and fine-tuned deep neural networks, optimizing hyperparameters to improve accuracy and mitigate overfitting; evaluated performance with F1-score, RMSE, and confusion matrices.

Autonomous Deep Q-Network (DQN) Agent

DS 510 – Artificial Intelligence for Data Science

- Architected and trained an autonomous reinforcement-learning agent in PyTorch to navigate a dynamic PyPacman environment, formalized as an MDP with a custom OpenAI Gym-compatible interface and multi-objective reward shaping.
- Engineered and evaluated Standard, Dueling, and Convolutional DQN architectures; stabilized training over 5,000 episodes using experience replay buffers, target networks, and epsilon-greedy exploration.
- Extracted spatial features (Manhattan distance, adjacent threat counts) to improve state-action mapping and overall decision-making efficiency.

eCommerce Web Application with Local AI (Gemma)

CS 628 – Full-Stack Web Application Development

- Developed a full MERN stack application (MongoDB, Express.js, React, Node.js) with an Ollama / Gemma 2 AI-powered chat assistant for product recommendations and user assistance.
- Built dynamic React UI components (browsing, search, cart, checkout), RESTful APIs for product management and order processing, and a modular architecture with structured IEEE documentation.

Education

M.S., Computer Science (AI/ML focus) — City University of Seattle *January 2025 – Present*

B.S., Computer Engineering & Systems — University of Washington Tacoma, June 2017 *Minor in Mathematics*